

花老师班补充作业1.

61. (1) $\frac{\sqrt{n+2\sqrt{n}}}{\sqrt{n} + \sqrt{n+2\sqrt{n}}} \rightarrow \frac{1}{2}$.

(2) $1 < \sqrt[n]{\sum_{k=1}^n \frac{1}{k}} < \sqrt[n]{n} \rightarrow 1$

64. $x_{n+1} = 1 + \frac{x_n}{1+x_n} \Rightarrow 1 < x_n < 2$.

$x_1 > 0$

$x_{n+1} - x_n = \frac{x_n}{1+x_n} - \frac{x_{n-1}}{1+x_{n-1}} = \frac{x_n - x_{n-1}}{(1+x_n)(1+x_{n-1})}$

$x_2 - x_1 = 1 + \frac{x_1}{1+x_1} - x_1 = \frac{-x_1^2 + x_1 + 1}{1+x_1} = \frac{-(x_1^2 - x_1 - 1)}{1+x_1} = \frac{-(x_1 - \frac{1+\sqrt{5}}{2})(x_1 - \frac{1-\sqrt{5}}{2})}{1+x_1}$

$1+x_1 > 0$

$\Rightarrow$ if $1 < x_1 < \frac{\sqrt{5}+1}{2} \Rightarrow x_2 > x_1$. Assume $x_n > x_{n-1} \Rightarrow x_{n+1} > x_n \Rightarrow \{x_n\}$ increases.
Since $1 < x_n < 2$
 $\Rightarrow \{x_n\}$ converges.

let $\lim_{n \rightarrow \infty} x_n = l \Rightarrow l = \frac{\sqrt{5}+1}{2}$

if $\frac{\sqrt{5}+1}{2} < x_1 < 2 \Rightarrow x_2 < x_1$. Assume $x_n < x_{n-1} \Rightarrow x_{n+1} < x_n \Rightarrow \{x_n\}$ decreases.

Since $1 < x_n < 2$
 $\Rightarrow \{x_n\}$ converges. let $\lim_{n \rightarrow \infty} x_n = l \Rightarrow l = \frac{\sqrt{5}+1}{2}$

16. $a_{n+1} - 9 = 3 - \sqrt{a_n} = \frac{9 - a_n}{3 + \sqrt{a_n}} \Rightarrow \left| \frac{a_{n+1} - 9}{a_n - 9} \right| = \frac{1}{3 + \sqrt{a_n}} < \frac{1}{3}$

$\Rightarrow \left| \frac{a_2 - 9}{a_1 - 9} \cdot \frac{a_3 - 9}{a_2 - 9} \cdots \frac{a_n - 9}{a_{n-1} - 9} \right| < \frac{1}{3^{n-1}} \Rightarrow 0 \leq |a_n - 9| < \frac{|a_1 - 9|}{3^{n-1}} = \frac{6}{3^{n-1}} \rightarrow 0$

$\Rightarrow \lim_{n \rightarrow \infty} a_n = 9$.

$$17. a_n = \frac{1}{2} \left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right) = \frac{1}{2} \left(\frac{1}{n} - \frac{1}{n+1} \right) - \frac{1}{2} \left(\frac{1}{n+1} - \frac{1}{n+2} \right)$$

$$\Rightarrow S_n = \frac{1}{2} \left(1 - \frac{1}{n+1} \right) - \frac{1}{2} \left(\frac{1}{2} - \frac{1}{n+2} \right) \rightarrow \frac{1}{4}$$

$$18. f(x) = \frac{1}{x^p} \Rightarrow \int_2^{\infty} \frac{dx}{x^p} = \int_{\ln 2}^{\infty} \frac{du}{u^p} \Rightarrow p > 1$$